

Classification of EEG for Upper Limb Motor Imagery: An Approach for Rehabilitation

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Abstract—Patients suffering from severe motor neuron diseases (*MND*) experience motor disability and their rehabilitation has always remained a challenge. Electroencephalogram (*EEG*) based brain computer interface (*BCI*) is a system that can be used for the rehabilitation of patients suffering from amputation or from severe disease like *MND*, stroke, locked in syndrome (*LIS*); where *EEG* signal is acquired from brain scalp while performing certain mental task such as motor imagery, cognitive imagery etc. In the present paper brain signals i.e. *EEG* for 10 motor imagery movements of upper limb acquired from 4 subjects were classified. Filter Bank Common Spatial Pattern (*FBCSP*) algorithm was used for extracting features of *EEG* signal captured from 5 electrodes placed over motor cortex and mutual information is used for feature selection. Classification algorithm followed was linear Support Vector Machine (*SVM*) in *MATLAB* 2015a.

Keywords: *Motor Imagery, EEG, MND, FBCSP, Mutual Information, Node, SVM*

I. INTRODUCTION

Brain is the most complex part of human body which continuously processes the information it receives from different sensory organs and is responsible for controlling the movement of output channels such as motor and muscles movements. Any message or command is firstly generated in the brain in the form of electrical signals (by firing of neurons) which is then sent to the output channel via nerves and spinal cord. Any blockage in the path of these signals can hinder the functioning of the muscles making a person disable. Severe muscular disease such as amyotrophic lateral sclerosis (*ALS*), progressive bulbar palsy (*PBP*) etc can degrade or degenerate the natural pathways of signal communication to the muscles making voluntary movement difficult, and as *MND* are progressive diseases the condition of the patients gets worsen day by day. According to world disability report of *WHO* (World Health Organisation) around 1 billion (15%) of the world population is suffering from some form of disability of which 2 – 3 % suffers severe disability of functions[1]. In India itself, over 21 million (2.1% of the population) people suffers from some kind of disability from which 28% facing disability in movement and 7.5% in speech[2].

Rehabilitation of patients suffering from disabled motor movements has always been a challenge as it requires personal care moreover traditional physical therapy provided by

therapist is expensive and time consuming. *BCI* is an alternate approach to the conventional therapy for the patients whose brain functions normally but communication pathways has been degraded due to neurological disorder or amputation. *BCI* harnesses the brain signals generated during actual or imagery motor movement which are then used as command signal to control an external hardware such as neuro prosthetics and orthotics etc. Here in this study *EEG* based *BCI* is used for motor imagery which can be used for rehabilitation of the patients suffering from severe impairment such as stroke, *LIS*, *MND* or amputation etc.

Experiments for decoding motor imagery movements of upper limb and fine movements as of fingers had been performed previously where actual and/or imagined movements were used. In 2008 Vuckovic et al. classified extension and flexion of left and right hand wrist using *EEG* recording of 64 channels achieving an accuracy of 69%[3]. They proposed a 2 stage 4 class classifier model which is the inspiration of working model used in this study and used amplitude of gabor coefficients as a feature to train a recurrent neural network classifier. In [4] authors decode individual finger movement of one hand using *EEG* signals from 128 electrodes using power spectral density (*PSD*) as a feature and principal component analysis (*PCA*) as the feature reduction technique. They used *SVM* for classifying ten pair of finger movement and achieved average decoding accuracy of 77.11%. In another study in 2017 Minmim Miao et al distinguished between the movement imagery and rest state of the right hand index finger using band power and approximate entropy as the features[5]. They used the proposed dictionary cleaned sparse representation-based classification, with *kNN* as a classifier and claimed to achieve average accuracy of 81.32% for 5 subjects. In [6] authors tried to discriminate between the movements of individual fingers with the help of *EEG* signal where movement of all fingers except the ring was considered. They conducted experiments for actual and imagined movement of individual fingers and achieved average classification accuracy of 61.8% for each pair of finger using *LDA* as a classifier.

In consideration of the discussed literature, here we tried to discriminate between the *EEG* signals acquired during imagery of 10 motor movements of upper limb which, can be used for the rehabilitation of patients suffering from disability such as paretic hands, monoplegia, hemiplegia etc. For the

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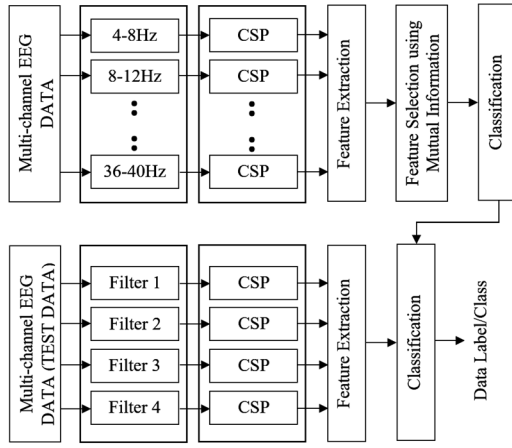


Fig. 3 Architectural flow chart for FBCSP

- CSP- introduced by Zoltan J. Koles et al in 1990 and since then is being used in detection of event related desynchronisation (*ERD*)/ event related synchronisation (*ERS*) in *EEG* as it is highly successful in finding the spatial filters [8-11]. It is a mathematical feature extraction technique used to discriminate between 2 classes by using a transformation matrix that increases the variance of one condition while simultaneously minimizing it for other. The method used in finding the transformation matrix is based on simultaneous diagonalization of two matrices (covariance matrices of two classes) and can be calculated by calculating Eigen values for two class problem[12]. The normalized covariance matrices are given by (1).

$$R_i = X_i * X_i' / \text{trace}(X_i * X_i') \quad \text{for } i = 1, 2 \quad (1)$$

And spatially filtered signal is given as,

$$Z_i = W X_i \quad (2)$$

Where X is *EEG* signal with dimension of $C \times T$, C is the number of channels, T is the number of sample points in a trial and W is the CSP projection/transformation matrix. The column of W^{-1} are the common spatial pattern and row of W are stationary spatial filters[7]. The transform signal given by (2) used for feature extraction as it maximizes the difference in the variance of two classes of *EEG* signal. However, only the first and last m rows of the spatially filtered signal are usually considered as they have maximum difference in the variance. The CSP features that were used is given by (3)

$$f_v = \log \left(\frac{\text{var}(Z_v)}{\sum_{v=1}^{2m} \text{var}(Z_v)} \right) \quad (3)$$

where $v = 1 \dots 2m$

The value of m used in this study is 1 i.e. first and last row of the spatially transformed *EEG*.

- Feature Selection- is the process of removing redundant or irrelevant features from original set of features such that loss of information is minimal. It is very useful tool as it decreases the dimensionality of features which in turns reduces the computational cost, time and also makes the classifier more robust by avoiding over fitting. There are mainly three approaches that are used in feature selection; wrapper method, filter method and embedded method of feature selection. In wrapper method features are selected on the basis of classifier, where classifier is trained and tested for different subset of features due to which it is computationally heavy. In filter method feature subset is selected independent of the classifier, it ranks the features based on their statistical dependence with each other. Embedded methods are hybrid methods in which feature selection is done for a given classifier while feature subset is chosen using another learning algorithm itself as a part of evaluation function[13]. In this study, filter method for feature selection was used and the features were selected on the basis of mutual information[14, 15]. Feature ranking was done on the basis of mutual information that they carry given by (4), the larger the value of mutual information means the greater predictive ability the feature holds.

$$I(X, Y) = H(Y) - H(Y/X) \quad (4)$$

Where x and y are two random variable and $H(Y)$ is the entropy given as,

$$H(Y) = \sum_{y \in Y} p(y) \log_2 p(y) \quad (5)$$

And conditional entropy of X and Y $H(Y/X)$ is given as,

$$H(Y/X) = \sum_{x \in X} \sum_{y \in Y} p(x, y) \log_2 p(y/x) \quad (6)$$

The first 8 features out of whole feature set given by feature selection algorithm sorted in decreasing order of relevance were selected to train the classifier. Since CSP gives 2 features (variance of first & last row of spatially transformed *EEG*) therefore 4 feature pairs were selected, making a total of 8 features.

- Classification: *SVM* with linear kernels was used as the classification algorithm at each node. *SVM* performs classification by constructing a hyper plane in high dimension feature space and finds support vectors that maximize the distance between the nearest data point/feature of each class and the hyper plane. At each node 75% of the data from each class was used for training and rest 25% was used for testing the train model. The work flow related to the classification is shown in fig. 4.

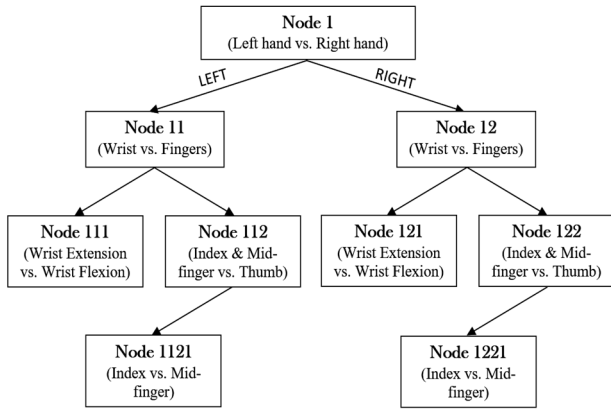


Fig. 4 Work flow of classification model followed.

III. RESULTS AND DISCUSSION

The results obtained using linear *SVM* classification at each node is shown in table 1 for each subject. Classification of imagery movements is done by dividing 10 class problem into several binary class problems to improve the recognition rate. At each end node the classification of *EEG* data of one of the 10 motor imagery movements has been performed using linear *SVM* for which 9 parallel classifiers were used.

Table 1: Recognition rate achieved at each node.

Subjects Node	1	2	3	4	AVG
1	100	100	100	98	99.5
11	100	100	100	100	100
111	100	100	80	80	90
112	100	100	100	90	97.5
1121	100	100	100	100	100
12	81.67	75	95	100	87.92
121	80	100	100	100	95
122	100	85	65	90	85
1221	100	90	100	70	90

In first layer at node 1 the classification of 2 classes one being left hand 'L' and the other right hand 'R' (i.e. all the movements of right hand v/s all the movements of left hand) was performed. In second layer further two classifications at node 11 and node 12 were performed where the former one classifies between the left hand wrist movements (i.e. left wrist extension 'LWE' & left wrist flexion 'LWF') and finger movements (i.e. left Index finger movement 'LIF', left middle finger movement 'LMF' & left Thumb movement LTH) and latter one classifies between right hand wrist (i.e. right wrist extension 'RWE' & right wrist flexion 'RWF') and finger movements (i.e. right index finger movement 'RIF', right middle finger movement 'RMF' & right thumb movement RTH). Further in layer third 4 classifications at node 111 and node 112 (for left hand) and node 121 and node 122 (for Right hand) were done. Where at node 111 classification between LWE v/s. LWF and at node 112 classification between LIF & LMF (two classes combined) v/s. LTH was

performed. In similar manner at node 121 (RWE v/s. RWF) and 122 (RIF & RMF v/s. RTH) classifications of right hand movements were performed. In last layer at node 1121(LIF v/s. LMF) and at node 1221 (RIF v/s. RMF) classification for left and right hand respectively were performed.

Comparison of recognition rate achieved for each class of all subjects can be seen in fig. 5 where horizontal line shown, represents the average performance (i.e. 91.5%) of the model. Also, it can be seen that the lowest average recognition rate achieved per class is 75% which is at node 122 as there is data unbalance during training (i.e. training sample of one class is more when compared to the other class).

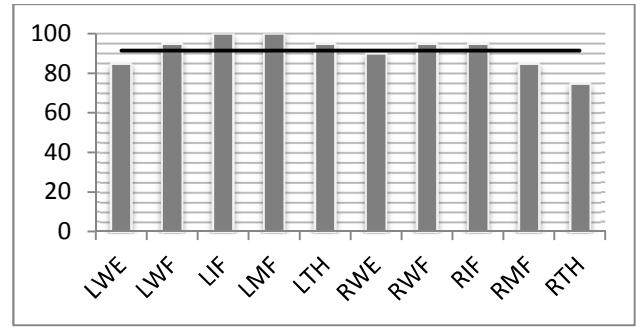


Fig. 5 Average recognition accuracies achieved for LWE, LWF, LIF, LMF, LTH, RWE, RWF, RIF, RMF and RTH imagery movements.

Average accuracy achieved per subject is shown in fig. 6 where it can be seen that subject 2 gave maximum accuracy while accuracy achieved by subject 4 is lowest. The origin of finger movement's activation in brain is same, as there is less spatial distance in between that's why it is difficult to distinguish them, this can be seen from table 1 where the accuracy achieved is less at node 1221 for classification of index and middle. It could also be possible due to the induced mental fatigue while performing imagery during prolonged acquisition session as reported by subjects.

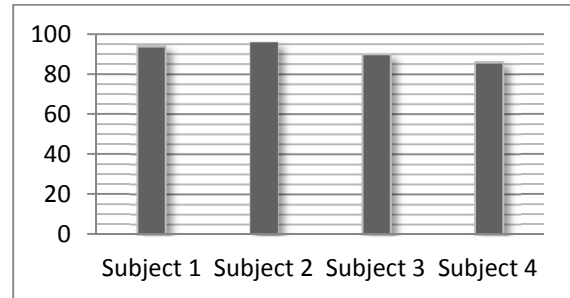


Fig. 6 Average recognition accuracies achieved per subject for all classes

This research was done using medical grade equipment and uses only 5 channels of *EEG* signals positioned over motor cortex. The classes/motor movements considered here are 10, giving a chance accuracy of 10% while the recognition rate achieved in this study is far better than that.

IV. CONCLUSION AND FUTURE SCOPE

The results achieved in this study are satisfactory and shows the feasibility of *EEG* based *BCI* rehabilitation devices for upper limb. As the 10 class problem is divided into several

binary problems, its success rate is increased moreover the lower accuracy is received either where there is data unbalance in training or where the source of generating potential is same (i.e. finger movements). Also the current study is limited to normal subjects, the viability of the methodology discussed here can be tested on patients suffering from disability so that hardware for the rehabilitation of upper limb can be designed using this methodology.

This research work can be further generalized by increasing the sample size and taking data from multiple volunteers and regularizing it for others. This work in future can be extended by adding more movements of wrist such as radial deviation, ulnar deviation, pronation & supination and more finger movements.

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